

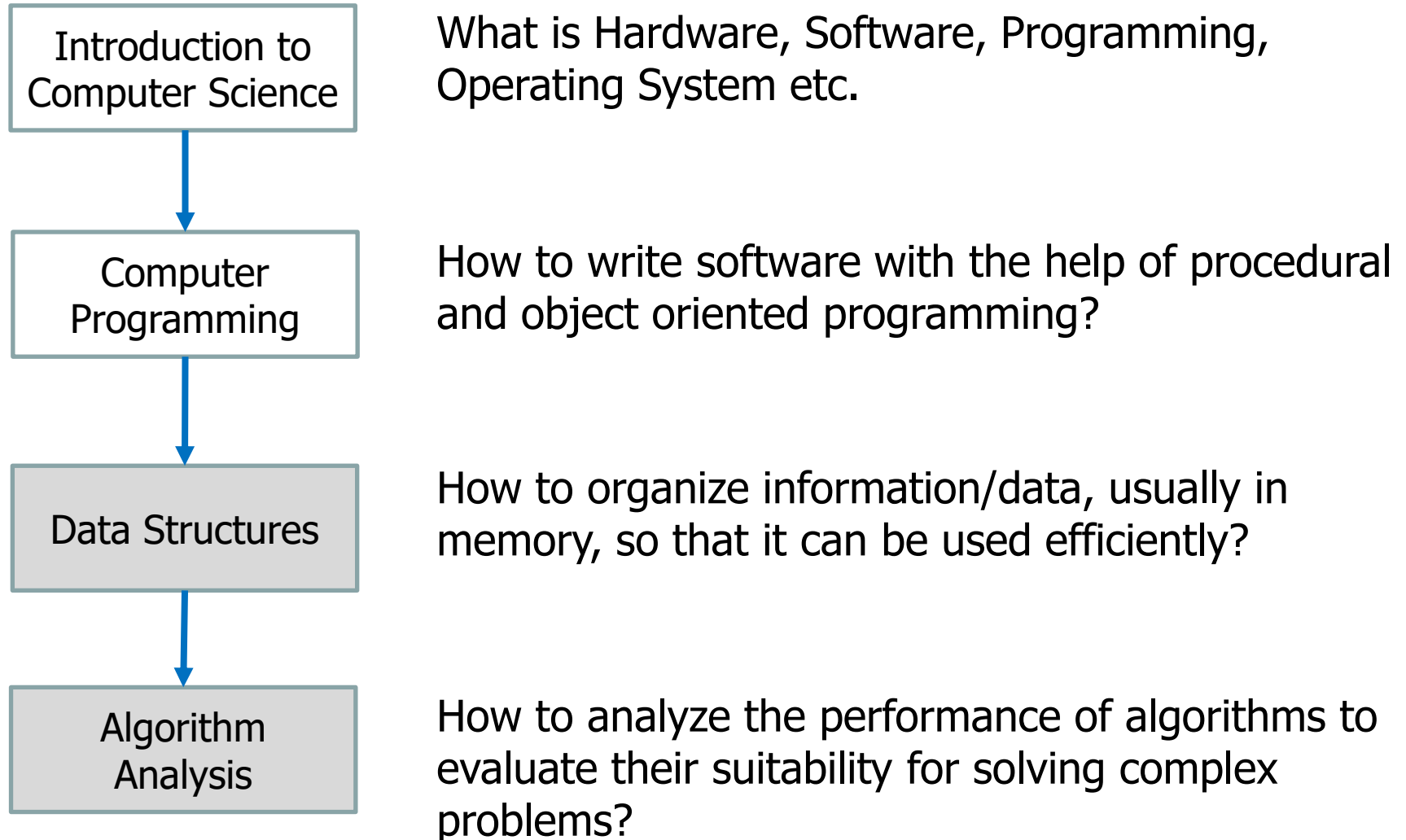
Data Structures

Fall 2024

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National University of Computer and Emerging
Sciences

General Overview



Some Rules

- Raise your hand before asking any question
 - Wait for the permission
 - Never ever miss a class
 - No retakes (except for Final Exam*)
 - Never use mobile phone in the class
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- Above all, whatever you do, please do not disturb others

* Conditional: as per university policy

Dishonesty and Plagiarism

- Any kind of cheating will be considered serious offense
- All parties involve in cheating will get minus (-) 50% marks
 - Exams, quizzes, assignments, projects
- Habitual cases will be awarded "F"

Attendance Policy

- Student arriving after the attendance will be marked “late”
 - No excuse to students arriving immediately after the attendance
- Two late arrivals will be treated as absent
- Students are not allowed to switch sections

What is a Data Structure?

- In a general sense, any representation that is used for storing information is a data structure
 - Example: integers, structures, classes, arrays, etc.
- More typically, a **data structure** provides **a way for organizing a collection of data items**
 - Storing, accessing and modifying data items
- Most data structures have associated **algorithms to perform operations** on data
 - Search, insert, remove etc.

Costs and Benefits

- Each data structure requires
 - **Space** for storing data items
 - **Time** to perform each basic operation
 - Programming **efforts**
- **Cost** of a data structure
 - Time and space resources in consumes
- Choice of data structure depends on many factors
 - **Type** of data
 - **Frequency** with which various **data operations** are applied
- Hard to define data structure that performs better in all situations
 - Time and space tradeoff

Goals of This Course

- Learn the commonly used data structures
 - Form a programmer's basic data structure "toolkit"
- Case studies of data structures
- Examine the costs and benefits of every data structure

Course Content

- Simple and abstract data types
- Arrays
- Background: Templates and Algorithm Analysis
- Searching and sorting techniques
- Linked Lists
- Queues
- Stacks
- Trees: Definitions and terminology

Mid term

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- Trees: Binary Search Trees, AVL Trees
 - Heap
 - Graphs
 - B+ Trees
 - Hashing

Tentative Evaluation Breakdown

Evaluation Name	Weightage
Quiz	5
Lab	10
Assignment	15
Project	10
Midterm	20
Final	40

- Labs will be conducted from week 2 onwards
 - Labs attendance is mandatory!
 - Hard to get good marks in assignments without lab

Motivational Example

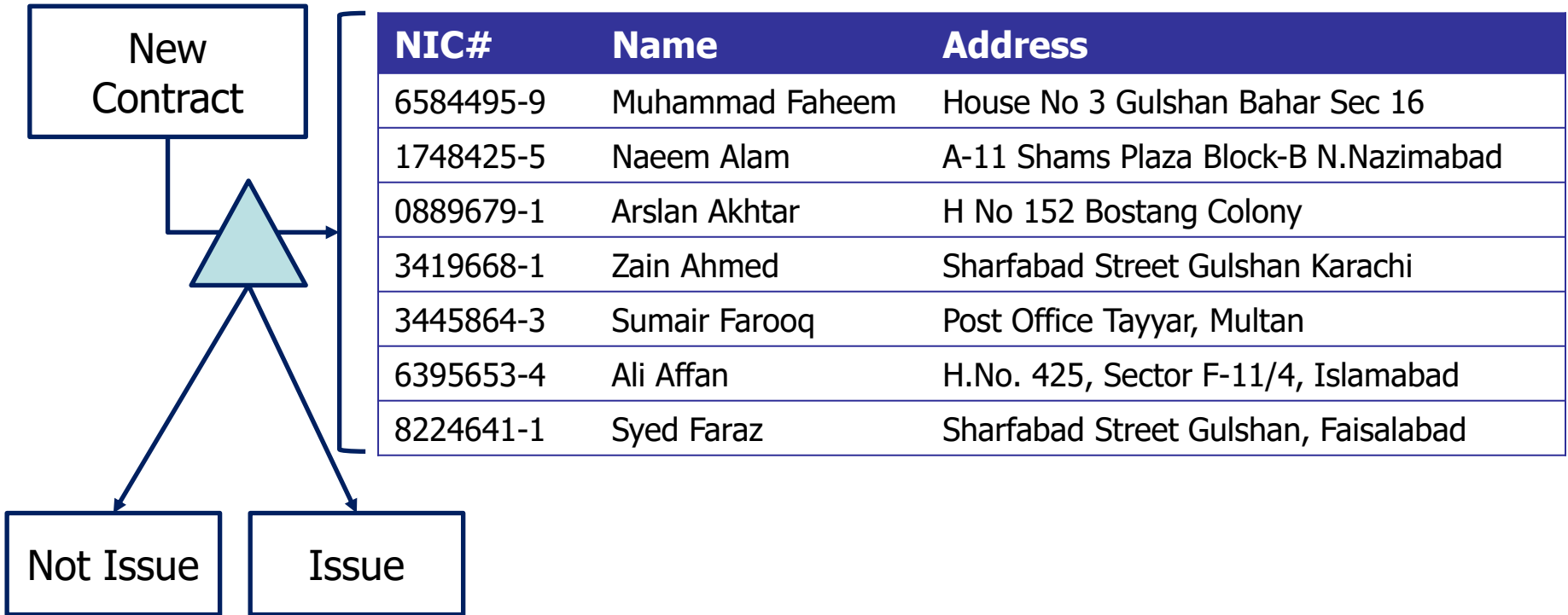
- A cellular service company provides contract to its 10 million users
- Due to new security enforcements, the company wants to **prevent issuing of multiple contracts to users**
- Method of Detecting Multiple Contracts
 - Before issuing a new contract to user
 - First search the **id of user** in existing contracts database
 - In case of failure → issue a new contract
 - In case of success → do not issue a new contract to user

Example: Linear Array Data Structure (1)

NIC#	Name	Address
6584495-9	Muhammad Faheem	House No 3 Gulshan Bahar Sec 16
1748425-5	Naeem Alam	A-11 Shams Plaza Block-B N.Nazimabad
0889679-1	Arslan Akhtar	H No 152 Bostang Colony
3419668-1	Zain Ahmed	Sharfabad Street Gulshan Karachi
3445864-3	Sumair Farooq	Post Office Tayyar, Multan
6395653-4	Ali Affan	H.No. 425, Sector F-11/4, Islamabad
8224641-1	Syed Faraz	Sharfabad Street Gulshan, Faisalabad

- Linear Array (with 10 million entries)
 - 3 arrays (NIC, Name, Address)
 - Structured array
 - Class's object array

Example: Linear Array Data Structure (2)



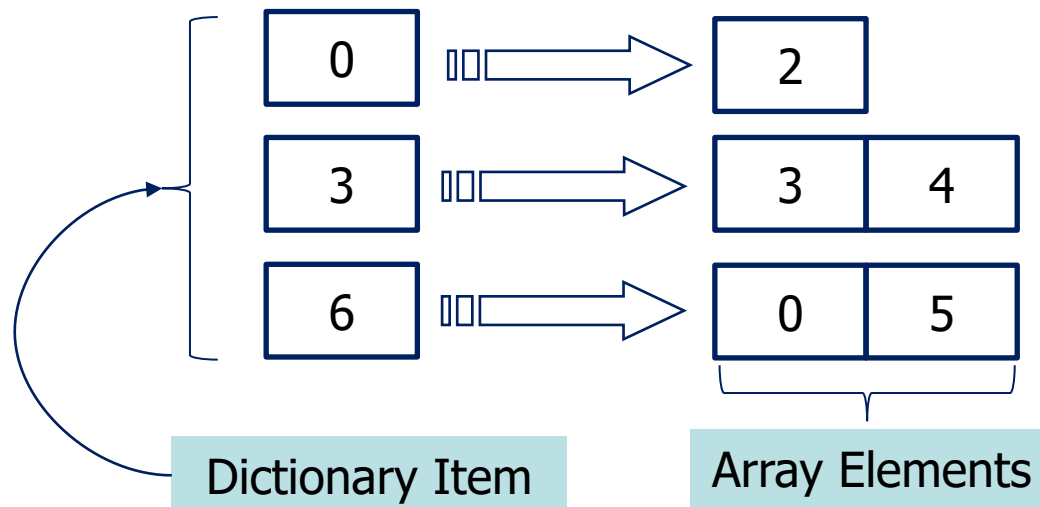
- Any disadvantage of linear array data structure?
- How to improve?

Example: Improved Data Structure

- Create a dictionary data structure
 - Group similar records together
 - Similarity in terms of first digit of NIC number
 - Add a dictionary entry for each distinct digit (0 – 9)
- Example: 3419668-1, 3445864-3, 1748425-5.
 - 3 and 1 are dictionary entries
- Existing contracts are searched in two steps
 1. Search the dictionary entry (i.e., group searching)
 2. Search contract within the group (i.e., with the same NIC digit)
- How much improvement w.r.t. linear array?

Example: Dictionary Data Structure (1)

NIC#	Name	Address
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1748425-5	Naeem Alam	A-11 Shams Plaza Block-B N.Nazimabad
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Example: Dictionary Data Structure (2)

- Another possibility
 - Maintain pointers with structures (or records)
 - Non NULL pointer indicates presence of next record

	NIC#	Name	Address
0	6584495-9	Muhammad Faheem	House No 3 Gulshan Bahar Sec 16
3	1748425-5	Naeem Alam	A-11 Shams Plaza Block-B N.Nazimabad
6	0889679-1	Arslan Akhtar	H No 152 Bostang Colony
	3419668-1	Zain Ahmed	Sharfabad Street Gulshan Karachi
	3445864-3	Sumair Farooq	Post Office Tayyar, Multan
	6395653-4	Ali Affan	H.No. 425, Sector F-11/4, Islamabad
	8224641-1	Syed Faraz	Sharfabad Street Gulshan, Faisalabad

Course & Reference Books

- Data Structures Using C++ (By D.S. Malik)
- Data Structures Using C and C++ (By Y. Langsam, M. J. Augenstein, A. M. Tenenbaum)
- Data Structures and Algorithms (By A. V. Aho, J. E. Hopcroft, J. D. Ullman)
- Schaum's Outline Series, Theory and problems of Data Structures (By Seymour Lipschutz)

Any Question So Far?



0			
3			
6			
	NIC#	Name	Address
	6584495-9	Muhammad Faheem	House No 3 Gulshan Bahar Sec 16
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Data Structure

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Where data structure is helpful?

- Data structure can make the difference between a program running in a few seconds or many days
 - Efficiency of data structure

What is Data Structure Efficiency?

- Two major measures of efficiency of a data structure or algorithm
 - Time and space
- Cost of data structure
 - Time and space resources it consumes
- Efficiency of a data structure depends on its operations
 - Makes a difference between program running in few seconds or days
- Difficult to define data structure that performs better in all situations